XMAKE • GRAPHICS ON METAL

丝网印刷
Silk Screen Printing

中英文对照审核稿 / Bilingual Review Draft

Primary keyword

silk screen printing metal

Content preview · Not the publish-ready WordPress version

内容预览 · 非最终 WordPress 发布版

Review Status：中英文对照审核稿 / Bilingual Review Draft
Suggested URL：`/silk-screen-printing-metal/`
Canonical URL Status：待确认现有站点结构 / Confirm against the existing site structure
Meta Title：Silk Screen Printing Metal Parts | Custom Graphics - Xmake
Meta Description：Xmake 提供 metal parts 丝网印刷服务，用于面板文字、Logo、符号、刻度和多色图形。项目评审覆盖金属表面状态、油墨体系、图稿分色、套准、固化及耐久性要求。
Meta Description: Xmake provides silk screen printing for metal parts, including panel legends, logos, symbols, scales, and multicolor graphics. Project review covers metal surface condition, ink system, color separation, registration, curing, and durability requirements.
Primary Keyword：silk screen printing metal
Secondary Keywords：silk screen printing on metal, metal screen printing services, custom metal printing, screen printing aluminum, screen printing stainless steel, printed metal panels, industrial screen printing
Query Variants：Can you screen print on anodized aluminum? · Which ink is used for metal screen printing? · Screen printing vs laser marking on metal · How durable is screen printing on metal? · What artwork is needed for metal screen printing?
Key Entities：screen mesh · stencil · squeegee · ink · substrate · surface energy · color separation · registration · curing · cross-cut test
Search Intent：交易型 / 商业调研型；买家正在确认金属表面与油墨是否兼容，并准备提交面板文字、Logo、符号、颜色、位置和耐久性要求。
Search Intent: Transactional / commercial investigation; the buyer is checking metal-surface and ink compatibility and preparing requirements for panel legends, logos, symbols, color, placement, and durability.

H1：金属零件丝网印刷 / Silk Screen Printing for Metal Parts

一批控制面板在出货检验时颜色正确、位置也正确，安装后却被清洁剂擦花。这样的失效通常不是“印刷不够牢”一句话能解释：金属实际表面、油墨化学体系、固化程度和清洁介质必须放在一起判断。

Xmake 提供 silk screen printing metal 服务，用于把文字、Logo、操作符号、刻度和色块印到定制金属零件上。丝网印刷适合平面和经评估的缓曲面，可形成清晰、较高遮盖力的图形；项目结果取决于图稿、网版、油墨、表面准备、工装、套准和固化。

A batch of control panels can pass shipment inspection for color and position, then smear when cleaned after installation. That failure cannot be explained by saying the print was simply “not durable enough”: the actual metal surface, ink chemistry, cure state, and cleaning medium must be evaluated together.

Xmake provides silk screen printing for metal parts, placing text, logos, operating symbols, scales, and solid color areas on custom components. Screen printing suits flat and evaluated gently curved surfaces and can produce crisp, relatively opaque graphics; the result depends on artwork, screen, ink, surface preparation, tooling, registration, and cure.

H2：快速判断：这个图形适合丝网印刷吗？ / Quick Fit Check: Is Screen Printing the Right Process?

项目条件	丝网印刷通常合适	先比较其他工艺
图形内容	固定文字、Logo、警示符号、刻度、实色块	每件不同的序列号、二维码或实时可变数据
零件表面	平面、浅曲面或可稳定支撑的局部印刷区	深凹、强复合曲面、锐利台阶或网版无法靠近的区域
颜色需求	专色、较高遮盖力、少量到多色套印	照片级连续色调或频繁更换的小批图稿

项目条件	丝网印刷通常合适	先比较其他工艺
批量逻辑	同一图稿重复生产，制版成本可被批量分摊	单件试制且图稿仍持续变化
耐久要求	可以通过油墨、固化和样件测试建立验收	需要标记进入基材、不能依赖表面油墨层

Project Condition	Screen Printing Usually Fits	Compare Another Process First
Graphic content	Fixed text, logos, warning symbols, graduations, and solid areas	Unique serial numbers, QR codes, or live variable data on every part
Part surface	Flat, gently curved, or locally printable areas with stable support	Deep recesses, compound curves, sharp steps, or areas the screen cannot approach
Color need	Spot colors, relatively high opacity, and registered prints from one to several colors	Photographic continuous tones or frequently changing short-run artwork
Volume logic	Repeated production of one design can absorb screen setup	One-off prototypes while the artwork is still changing
Durability need	Acceptance can be built through ink, cure, and sample testing	The mark must enter the substrate and cannot rely on a surface ink film

结论 / Decision

固定图稿、平面金属件和清晰专色是丝网印刷的典型组合。变量数据优先比较激光打标或数字印刷；小面积复合曲面优先比较移印；需要凹入永久标记时优先比较 CNC 或激光雕刻。

Fixed artwork, flat metal parts, and clear spot colors are a typical screen-printing combination. Compare laser marking or digital printing for variable data, pad printing for small compound curves, and CNC or laser engraving when the mark must be recessed into the substrate.

H2：丝网印刷如何把图稿转移到金属上 / How Screen Printing Transfers Artwork to Metal

网框上的张紧网布被感光材料制成图文模板，图形区域保持开放。刮板推动油墨通过开放网孔，油墨在网版离开后留在金属表面，随后按油墨体系进行挥发干燥、化学交联、热固化或 UV 固化。

“Silk screen” 是沿用至今的工艺名称，不表示现代网版必须使用天然丝。实际网布、目数、张力、乳剂厚度、离网距离和刮板条件会共同影响墨层厚度、边缘清晰度和细节再现。

A tensioned mesh on a frame is converted into an image stencil with open graphic areas. A squeegee drives ink through those openings; the ink remains on the metal after the screen lifts and is then dried by solvent evaporation, chemical crosslinking, thermal cure, or UV cure according to the ink system.

“Silk screen” is the established process name and does not mean a modern screen must use natural silk. Mesh type and count, tension, emulsion thickness, off-contact distance, and squeegee conditions work together to control ink deposit, edge definition, and detail reproduction.

H2：先写清表面状态，再写金属名称 / Specify Surface Condition Before Metal Name

“铝”或“不锈钢”不足以完成油墨选型。同一种铝件可能是裸铝、阳极氧化、喷粉、湿式喷漆或镀层表面，这些界面的润湿、溶剂敏感性、表面能和固化耐温完全不同。

“Aluminum” or “stainless steel” is not enough for ink selection. The same aluminum part may be bare, anodized, powder coated, wet painted, or plated, and each interface has different wetting, solvent sensitivity, surface energy, and cure-temperature limits.

实际印刷表面	主要风险	报价前需要的信息
裸铝	氧化层、加工油、指纹、表面粗糙度和批次状态影响润湿与附着	合金、加工状态、清洗、外观等级和是否需要前处理

实际印刷表面	主要风险	报价前需要的信息
阳极氧化铝	封孔状态、染色、光泽和表面处理供应批次会改变附着	阳极氧化类型、颜色、封孔、现有规范和实际样件
不锈钢	油污、钝化膜、抛光膏和低表面能污染物可能导致缩墨或脱落	牌号、表面等级、钝化 / 抛光状态和清洁限制
涂装金属	实际接触面是粉末涂层或油漆，不是下面的金属	涂层树脂类型、固化状态、光泽、添加剂、重涂窗口和耐温
镀层金属	镀层化学、孔隙、保护膜和防指纹处理会改变油墨结合	镀层类型、后处理、清洁限制和样件测试

Actual Printing Surface	Main Risk	Information Needed Before Quotation
Bare aluminum	Oxide, machining oil, fingerprints, roughness, and batch condition affect wetting and adhesion	Alloy, manufacturing condition, cleaning, appearance grade, and pretreatment need
Anodized aluminum	Sealing, dye, gloss, and finishing batches can change adhesion	Anodizing type, color, sealing, existing specification, and actual sample
Stainless steel	Oil, passive film, polishing compound, and low-surface-energy contamination can cause crawl or release	Grade, surface finish, passivation/polishing state, and cleaning restrictions
Coated metal	The ink contacts powder coat or paint, not the metal below	Coating resin, cure state, gloss, additives, recoat window, and temperature limit
Plated metal	Plating chemistry, pores, protective films, and anti-fingerprint treatments alter bonding	Plating type, post-treatment, cleaning restrictions, and sample testing

表面准备必须与零件状态相匹配。除油、除尘、去除指纹或经验证的表面活化可能有帮助，但未经确认的强溶剂、打磨或火焰处理也可能损伤阳极层、漆膜或外观面。

Surface preparation must match the part condition. Degreasing, dust removal, fingerprint removal, or a validated activation step may help, while an unapproved aggressive solvent, abrasion, or flame treatment can damage anodizing, paint, or a cosmetic surface.

H2：从矢量图稿到可印刷色版 / From Vector Artwork to Printable Separations

矢量主文件 / Vector Master

文字、线条、Logo 和刻度应优先提供可编辑矢量文件，并把字体转曲或同时提供受控字体。仅提供低分辨率截图会带来边缘锯齿、笔画变化和尺寸解释问题。

Text, lines, logos, and graduations should be supplied as an editable vector master, with fonts outlined or provided as controlled font files. A low-resolution screenshot introduces jagged edges, stroke changes, and ambiguous dimensions.

每色独立分色 / One Separation per Color

传统专色丝网印刷通常为每一种颜色制作独立色版，并逐色套印。颜色数量增加会带来更多网版、固化步骤和套准累积，图稿应明确叠印、镂空、底白和颜色顺序。

Traditional spot-color screen printing normally uses a separate stencil for each color and registers the colors in sequence. More colors add screens, cure steps, and cumulative registration variation, so artwork must define overprints, knockouts, white underbase, and print order.

深色底材上的遮盖 / Opacity on Dark Substrates

深色阳极氧化、黑色喷粉或彩色漆面会改变油墨视觉结果。浅色图形可能需要高遮盖配方、底白或多次印刷；增加墨层也会影响细节、固化和边缘，因此应以实际底材样件确认。

Dark anodizing, black powder coat, or colored paint changes the visual result. Light graphics may require a high-opacity formulation, white underbase, or multiple print passes; added ink deposit also affects detail, cure, and edges, so the actual substrate must be sampled.

尺寸由图纸控制 / Dimensions Controlled by the Drawing

印刷位置不能只靠效果图。图纸应给出图形相对于孔、边缘、中心线或装配基准的位置，并标明主视方向、允许偏移、禁印区和裁切边界。

Print position should not rely on a rendering alone. The drawing should locate the graphic from holes, edges, centerlines, or assembly datums and define viewing direction, allowed shift, no-print zones, and trim boundaries.

H2：四类金属丝印项目 / Four Metal Screen-Printing Programs

控制面板与操作文字 / Control-Panel Legends

适用于按钮名称、端口文字、警示符号、方向箭头和刻度。设计重点不是把字体做得尽可能小，而是保证安装方向、操作距离和最终表面反光条件下可读。

典型零件：铝面板 · 不锈钢面板 · 喷粉钣金件 核心输入：字体 / 符号 · 位置基准 · 颜色 · 清洁剂 · 可见面

Used for button names, port labels, warnings, direction arrows, and graduations. The design goal is not the smallest possible type; it is legibility at the installed orientation, operating distance, and final surface reflectance.

Typical Parts: Aluminum panels · Stainless-steel panels · Powder-coated sheet-metal parts Core Inputs: Type/symbol · Position datum · Color · Cleaner · Visible surface

Logo 与品牌图形 / Logos and Brand Graphics

Logo 印刷需要同时控制矢量轮廓、专色、底材颜色和商标留白区。金属色、荧光色或特殊效果油墨的视觉结果与普通专色不同，应先确认材料可得性、样板和批次接受方式。

典型零件：外壳 · 铭牌 · 盖板 · 装饰面板 核心输入：矢量 Logo · 色样 · 底材 · 光泽 · 观察条件

Logo printing requires control of vector outlines, spot color, substrate color, and clear space. Metallic, fluorescent, or other effect inks look different from standard spot colors and require availability, sample, and batch-acceptance review.

Typical Parts: Housings · Nameplates · Covers · Decorative panels Core Inputs: Vector logo · Color sample · Substrate · Gloss · Viewing condition

多色套印图形 / Registered Multicolor Graphics

多色项目按颜色顺序逐次印刷和固化，套准误差会在细边、反白文字和相邻色块处最明显。图稿可以使用合理的 trapping 或间隙设计降低视觉风险，但具体补偿需要结合颜色顺序和油墨流变确认。

典型零件：人机界面面板 · 多语言操作面板 · 彩色标识 核心输入：分色文件 · 套准容差 · 印刷顺序 · 底白 · 首件样

Multicolor work is printed and cured in sequence, making registration variation most visible around fine borders, reversed text, and adjacent color blocks. Practical trapping or clearance can reduce visual risk, but compensation must match print order and ink rheology.

Typical Parts: HMI panels · Multilingual operator panels · Color-coded signs Core Inputs: Separations · Registration tolerance · Print order · White underbase · First article

功能符号与工艺标记 / Functional Symbols and Process Marks

丝印可用于固定的极性、接地、连接、方向、量程或安全图形。若标记承担法规、操作安全或追溯功能，内容、位置、耐久性和版本控制必须由项目文件明确，不能把普通外观检查当作完整验证。

典型零件：电气外壳 · 仪器面板 · 固定规格铭牌 核心输入：受控符号 · 版本 · 使用环境 · 磨耗 / 清洁 · 法规要求

Screen printing can apply fixed polarity, grounding, connection, direction, range, or safety graphics. If the mark carries regulatory, operating-safety, or traceability meaning, content, position, durability, and revision control must be defined by project documents; cosmetic inspection alone is incomplete.

Typical Parts: Electrical housings · Instrument panels · Fixed-specification nameplates Core Inputs: Controlled symbol · Revision · Environment · Wear/cleaning · Regulatory requirement

H2：油墨选择从使用环境反推 / Select Ink Backward from the Service Environment

使用条件	油墨与固化评审重点	不应直接假设
室内低频触碰	基材附着、颜色、完全固化和普通擦拭	“室内”就不需要附着或清洁测试
频繁手触与按键区	皮脂、磨耗、摩擦抛光、清洁剂和图形位置	高光油墨一定比哑光更耐磨
户外暴露	UV、温湿循环、雨水、底材涂层和颜色保持	所有颜色和所有油墨具有相同户外寿命
化学品或消毒清洁	介质名称、浓度、接触方式、停留时间和循环次数	“耐酒精”代表耐所有溶剂或消毒剂
后续覆膜或透明层	层间附着、溶剂相容、颜色变化和总工艺窗口	透明层一定提高耐久且不会改变外观
热固化受限组件	零件、胶粘件、密封件和原有涂层的耐温	金属本体耐温就等于整个组件耐温

Service Condition	Ink and Cure Review Focus	Do Not Assume
Indoor, infrequent touch	Substrate adhesion, color, full cure, and routine wiping	“Indoor” eliminates the need for adhesion or cleaning tests
Frequent hand contact or keypad zone	Skin oils, abrasion, friction polishing, cleaner, and graphic location	Gloss ink always wears better than matte ink
Outdoor exposure	UV, temperature/humidity cycles, rain, substrate coating, and color retention	Every color and ink has the same outdoor life
Chemical or disinfectant cleaning	Chemical name, concentration, contact mode, dwell time, and cycles	“Alcohol resistant” means resistant to every solvent or disinfectant
Laminate or clear overprint	Intercoat adhesion, solvent compatibility, color change, and total process window	A clear layer always improves durability without changing appearance
Heat-limited assembly	Temperature limit of the part, adhesives, seals, and existing coatings	Heat tolerance of the metal equals heat tolerance of the full assembly

油墨系列必须通过技术数据表和实际基材样件选择。同一油墨在不同阳极氧化批次、粉末涂层树脂或防指纹表面上的表现可能不同；供应商列出“metal”也不能替代项目试印。

The ink series must be selected through its technical data sheet and an actual-substrate trial. One ink may behave differently across anodizing batches, powder-coat resins, or anti-fingerprint surfaces; a supplier listing “metal” does not replace a project print trial.

H2：几何和工装决定能不能印到位 / Geometry and Tooling Determine Print Placement

平面零件 / Flat Parts

平板、面板和铭牌最容易建立稳定支撑、离网和位置基准。凸起螺柱、已装配五金、翘曲或邻近折弯会限制网框与刮板路径，即使印刷区域本身看起来是平的。

Flat plates, panels, and nameplates provide the clearest support, off-contact, and location datums. Raised studs, installed hardware, warpage, or nearby bends can obstruct the screen or squeegee path even when the print zone itself appears flat.

浅曲面 / Gently Curved Surfaces

部分单向缓曲面可以通过适配网版、工装和印刷行程评估。曲率会改变网版接触、墨层和图形尺寸，不应把平面图稿直接包覆后假定线宽与位置不变。

Some shallow, one-direction curves can be reviewed with suitable screen, tooling, and print travel. Curvature changes screen contact, ink deposit, and image dimensions; a flat graphic should not simply be wrapped around the surface with unchanged line width and position assumed.

圆柱与复合曲面 / Cylinders and Compound Curves

圆柱面可能需要专用旋转丝印设备和与圆周同步的工装。小面积复合曲面、凹面或多方向曲率通常更适合移印；选择取决于图形面积、曲率、位置和批量，而不是只看零件外形名称。

Cylinders may require dedicated rotary screen equipment and tooling synchronized to the circumference. Small compound curves, concave surfaces, or multi-directional curvature are often better suited to pad printing; the decision depends on graphic area, curvature, position, and volume rather than the part name alone.

H2：印刷缺陷应按现象追原因 / Diagnose Print Defects from the Observed Failure

现象	优先检查	常见修正方向
缩墨、鱼眼或局部拒墨	油污、硅、指纹、表面能、清洁剂残留	纠正清洁和污染控制，验证表面活化与油墨润湿
边缘毛糙或锯齿	图稿质量、网目、乳剂、网版张力、离网和油墨流动	修正矢量文件，调整制版与印刷参数
细线断开或小字糊版	线宽、网孔开放、油墨黏度、刮板和网版释放	放宽细节或调整网版、油墨和刮印条件
套色偏移	定位基准、工装间隙、零件公差、网版伸长和固化顺序	改善基准与夹具，修正分色补偿和过程控制
擦拭掉色或划伤	基材附着、固化、墨层、清洁剂、磨耗载荷	更换或调整油墨体系，完成固化并按实际介质测试
色差或遮盖不足	底材颜色、墨层厚度、底白、批次和观察条件	用实际底材批准样件，控制配色、网版和印刷次数
气泡、针孔或表面不平	油墨混合、空气、底材粗糙、污染和固化挥发	调整混合与静置、印刷参数、表面准备和固化曲线

Symptom	Check First	Common Correction Direction
Crawling, fisheyes, or local ink rejection	Oil, silicone, fingerprints, surface energy, and cleaner residue	Correct cleaning and contamination control; validate activation and ink wetting
Ragged or sawtooth edge	Artwork quality, mesh, emulsion, tension, off-contact, and ink flow	Correct vector artwork and adjust screen-making and print parameters
Broken fine line or filled small text	Line width, mesh opening, ink viscosity, squeegee, and screen release	Open the detail or adjust screen, ink, and print conditions
Color misregistration	Location datum, fixture clearance, part tolerance, screen stretch, and cure sequence	Improve datums and fixture; revise separation compensation and process control
Wipes off or scratches	Substrate adhesion, cure, ink deposit, cleaner, and abrasion load	Change or adjust the ink system, complete cure, and test with actual media
Color shift or poor opacity	Substrate color, ink deposit, underbase, batch, and viewing condition	Approve on the actual substrate and control color mix, screen, and print passes
Bubbles, pinholes, or uneven surface	Ink mixing, entrained air, substrate texture, contamination, and cure volatiles	Adjust mixing/rest, print parameters, surface preparation, and cure profile

H2：Xmake 金属丝网印刷流程 / Xmake Metal Screen-Printing Workflow

1. 图稿与版本核对 / Artwork and Revision Check：确认矢量主文件、文字、Logo、颜色、分色、版本和受控符号。
2. 零件与表面确认 / Part and Surface Confirmation：核对金属牌号、实际表面处理、印刷区域、基准、曲率和已装配特征。
3. 油墨与测试计划 / Ink and Test Plan：根据颜色、清洁、磨耗、户外、化学和固化限制选择候选体系。
4. 分色与网版 / Separations and Screens：建立每色图层、印刷顺序、套准补偿、底白和网版参数。
5. 工装与表面准备 / Tooling and Surface Preparation：稳定零件位置，完成经验证的清洁，并保护非印刷区域。
6. 首件印刷 / First Article Print：检查内容、方向、位置、颜色、遮盖、细节、套准和表面缺陷。
7. 逐色印刷与固化 / Sequential Printing and Curing：按工艺顺序印刷各色，并控制中间干燥与最终固化。

8. 检验与保护包装 / Inspection and Protective Packaging：完成项目测试，防止零件相互摩擦、堆叠压印或包装材料粘连图形。

H2：丝网印刷与其他金属标记工艺 / Screen Printing vs Other Metal-Marking Processes

选择维度	丝网印刷	移印	激光打标	数字印刷
更适合的表面	平面和经评估的缓曲面	小面积复合曲面、凹凸面	光束可达且材料响应合适的表面	设备可处理的平面或专用三维工装
图形特点	专色、实色块、清晰文字和较高遮盖	小图形、局部符号、曲面转印	细小标识、序列号、二维码和基材对比	多色图像、短批和变量数据
生产准备	每色需要图稿分色、网版和套准	需要钢板 / 树脂版、胶头和工装	需要程序、参数和定位工装	需要色彩管理、文件和设备设置
耐久基础	取决于油墨-表面-固化组合	取决于油墨-表面-固化组合	取决于材料变化深度和后续环境	取决于油墨、底涂、固化和保护层
变量数据	不适合频繁逐件变化	不适合频繁逐件变化	通常适合	通常适合

Selection Dimension	Screen Printing	Pad Printing	Laser Marking	Digital Printing
Better-suited surface	Flat and evaluated gently curved surfaces	Small compound, concave, or convex areas	Beam-accessible surfaces with suitable material response	Flat surfaces or specialized 3D fixtures supported by the equipment
Graphic character	Spot colors, solid areas, clear text, and relatively high opacity	Small graphics, local symbols, and curved-surface transfer	Fine IDs, serial numbers, QR codes, and substrate contrast	Multicolor images, short runs, and variable data
Production preparation	Separations, screen, and registration for each color	Plate, pad, and tooling	Program, parameters, and location fixture	Color management, file, and machine setup
Durability basis	Ink-surface-cure combination	Ink-surface-cure combination	Depth or nature of material change and service environment	Ink, primer, cure, and protective layer
Variable data	Poor fit for frequent part-to-part changes	Poor fit for frequent part-to-part changes	Usually suitable	Usually suitable

H2：验收不能只看第一眼 / Acceptance Must Cover More Than First Appearance

验收项目	应在采购资料中定义	可使用的确认方式
内容与版本	受控图稿、语言、符号、零件版本和变更记录	与批准 PDF / 矢量文件逐项核对
位置与套准	基准、方向、位置公差、色间套准和禁印区	首件测量、透明叠图或项目检具
颜色与遮盖	实物色样、底材、光源、光泽和允许色差	同底材样件、受控目检或仪器测色
附着	实际表面、固化状态、测试时间和分级标准	项目适用的 ASTM D3359、ISO 2409 或批准方法
擦拭与耐化学	介质、浓度、擦拭材料、载荷、次数和判定	实际清洁剂或化学品循环测试
磨耗与触碰	接触物、载荷、路径、循环和允许变化	项目定义的摩擦 / 磨耗或模拟使用测试
外观缺陷	毛边、针孔、缩墨、划伤、漏印和可见面分级	标准光源下目检与限度样

Acceptance Item	Define in Purchasing Documents	Possible Confirmation Method
Content and revision	Controlled artwork, language, symbols, part revision, and change record	Line-by-line comparison with approved PDF/vector artwork

Acceptance Item	Define in Purchasing Documents	Possible Confirmation Method
Position and registration	Datums, orientation, location tolerance, color-to-color registration, and no-print zones	First-article measurement, transparent overlay, or project gauge
Color and opacity	Physical color sample, substrate, light source, gloss, and allowed difference	Same-substrate sample, controlled visual check, or instrumental color measurement
Adhesion	Actual surface, cure state, test timing, and rating requirement	Project-suitable ASTM D3359, ISO 2409, or approved method
Wipe and chemical resistance	Medium, concentration, wiping material, load, cycles, and acceptance	Actual-cleaner or chemical exposure cycle
Abrasion and touch	Contact material, load, path, cycles, and allowed change	Project-defined rub/abrasion or simulated-use test
Cosmetic defects	Ragged edge, pinhole, crawl, scratch, missing print, and visible-surface grade	Controlled-light inspection and limit samples

ASTM D3359-23 评估较有延展性的涂层在金属基材上经划切和胶带作用后的附着分级，不能给出绝对剥离强度。ISO 2409:2020 同样是划格抗分离评估；选择标准前应确认油墨膜、厚度、底材和项目规格是否在方法适用范围内。

ASTM D3359-23 rates relatively ductile coatings on metallic substrates after cutting and tape application; it does not provide an absolute peel-strength value. ISO 2409:2020 is likewise a cross-cut resistance-to-separation assessment. Before naming either standard, confirm that the ink film, thickness, substrate, and project specification fit the method scope.

H2：为什么由 Xmake 协调零件制造与印刷 / Why Coordinate Part Manufacturing and Printing Through Xmake

丝网印刷的位置、颜色和附着力都受上游制造影响。Xmake 可在钣金、CNC 加工、阳极氧化、喷粉或喷漆阶段就确定印刷基准、表面状态、工装空间和工序顺序，避免完成表面处理后才发现油墨或网版不适配。

- 图稿与零件版本绑定：受控文件对应正确的零件版本、语言和印刷方向。
- 表面处理与油墨联动：用实际阳极氧化、喷粉、漆膜或镀层样件做兼容性确认。
- 从首件到批量：首件同时核对外观、位置、套准、固化和项目定义的耐久性。
- 质量管理：Xmake 已通过 ISO 9001 和 ISO 13485 认证；这些是公司级质量管理资质，不代表具体印刷油墨自动满足法规、医疗、食品接触或安全标识要求。

Print position, color, and adhesion are affected by upstream manufacturing. Xmake can establish print datums, surface state, tooling clearance, and sequence during sheet-metal fabrication, CNC machining, anodizing, powder coating, or painting rather than finding an ink or screen conflict after finishing.

- Tie artwork to part revision: Controlled files correspond to the correct part revision, language, and print direction.
- Link finish and ink: Confirm compatibility on the actual anodized, powder-coated, painted, or plated surface.
- Carry first article into production: Check appearance, position, registration, cure, and project-defined durability together.
- Quality management: Xmake is ISO 9001 and ISO 13485 certified. These are company-level quality management credentials and do not mean that a specific ink automatically meets regulatory, medical, food-contact, or safety-marking requirements.

H2：常见问题 / Frequently Asked Questions

阳极氧化铝可以丝网印刷吗？ / Can Anodized Aluminum Be Screen Printed?

可以评估，但结果取决于阳极氧化类型、封孔、染色、表面污染和油墨体系。应使用实际量产表面试印；不能因为某一种阳极氧化样件通过，就推定所有颜色和供应批次都相同。

It can be evaluated, but the result depends on anodizing type, sealing, dye, contamination, and ink system. Print on the actual production surface; one successful anodized sample does not prove that every color and finishing batch will behave the same.

丝网印刷能做逐件变化的序列号吗？ / Can Screen Printing Produce Part-by-Part Serial Numbers?

技术上可以频繁更换网版或图稿，但通常不经济，也容易增加版本风险。逐件变量数据更适合激光打标或数字印刷，固定前缀和公共图形仍可保留丝印。

It is technically possible to change screens or artwork repeatedly, but it is usually uneconomical and increases revision risk. Laser marking or digital printing is better suited to part-by-part variable data, while a fixed prefix and common graphics can remain screen printed.

金属丝印需要透明保护层吗？ / Does Screen Printing on Metal Need a Clear Protective Layer?

不一定。适配并完全固化的油墨可能直接满足项目要求；透明层会增加一道材料界面，也可能改变颜色、光泽、附着和返修方式。是否使用应由磨耗、化学品、户外和外观测试决定。

Not always. A compatible, fully cured ink may meet the project requirements directly. A clear layer adds another material interface and can change color, gloss, adhesion, and repair behavior; use it only when wear, chemical, outdoor, and appearance testing support it.

最小文字和线宽是多少？ / What Is the Minimum Text or Line Width?

没有适用于所有网版、油墨和底材的统一最小值。字体笔画、阴阳字、网目、乳剂、墨层、底材粗糙度和套色数量共同决定可印性；应提交实际尺寸矢量图进行评估。

There is no universal minimum for every screen, ink, and substrate. Character stroke, positive or reversed text, mesh, emulsion, ink deposit, substrate roughness, and number of colors determine printability; submit full-size vector artwork for review.

询价需要提供哪些资料？ / What Should Be Included in an RFQ?

请提供 3D / 2D 零件文件、材料与实际表面处理、数量、矢量图稿、字体、颜色标准、印刷位置基准、套准要求、使用环境、清洁剂、磨耗条件和适用测试。若已有量产表面样件、色板或失效样，请一并提供。

Provide 3D/2D part files, material and actual surface finish, quantity, vector artwork, fonts, color standard, print-location datums, registration requirement, service environment, cleaner, wear condition, and applicable tests. Include production-finish samples, color standards, or failed parts when available.

CTA：获取金属丝网印刷项目评估 / Request a Metal Screen-Printing Review

上传零件图纸和矢量图稿，并说明金属实际表面、颜色、数量、位置、清洁方式和使用环境。Xmake 将评估制版、油墨、工装、套准、固化、首件和验收要求，再确认报价输入。

Upload the part drawing and vector artwork and specify the actual metal surface, color, quantity, location, cleaning method, and service environment. Xmake will review screen preparation, ink, tooling, registration, cure, first article, and acceptance requirements before confirming quotation inputs.

建议内部链接 / Suggested Internal Links

- Xmake Surface Finishing Services / Xmake 表面处理服务
- Sheet Metal Fabrication Services / 钣金加工服务
- CNC Machining Services / CNC 加工服务
- Anodizing Services / 阳极氧化服务
- Powder Coating Services / 粉末喷涂服务
- CNC Engraving Services / CNC 雕刻服务

- Request a Quote / 获取报价

人工核验说明 / Human Verification Notes

- Needs verification：实际印刷表面，包括阳极氧化、喷粉、油漆、镀层、钝化、防指纹或其他后处理的完整状态。
- Needs verification：油墨系列、颜色、混合 / 添加剂、适用网版、印刷次数、干燥 / 固化条件及可获得性。
- Needs verification：字体、最小线宽、阴阳字、颜色分版、底白、trapping、印刷顺序、套准和位置公差。
- Needs verification：清洁剂、化学品、磨耗、户外、温湿度、UV、触碰和包装接触条件及验收值。
- Needs verification：ASTM D3359、ISO 2409 或其他附着方法对实际油墨膜、厚度、底材和项目规格的适用性。
- Needs verification：法规符号、警示语、耐久性标签、食品接触、医疗、低 VOC、RoHS / REACH 或其他法定和客户专用要求。
- Needs verification: Complete state of the actual printing surface, including anodizing, powder coat, paint, plating, passivation, anti-fingerprint, or another post-treatment.
- Needs verification: Ink series, color, mix/additives, suitable screen, number of passes, drying/cure conditions, and availability.
- Needs verification: Font, minimum line width, positive/reversed text, separations, underbase, trapping, print order, registration, and position tolerance.
- Needs verification: Cleaner, chemical, abrasion, outdoor, temperature/humidity, UV, touch, and packaging-contact conditions and acceptance values.
- Needs verification: Applicability of ASTM D3359, ISO 2409, or another adhesion method to the actual ink film, thickness, substrate, and project specification.
- Needs verification: Regulatory symbols, warnings, durable labels, food contact, medical use, low VOC, RoHS/REACH, or other statutory and customer-specific requirements.

技术依据 / Technical Reference Basis

- `.agents/skills/xmake-plastichubs-website-ops/references/surface-finishing-knowledge-base.md`：支持丝网印刷用于平面或浅曲面上的图形、Logo、标签和面板，并要求按油墨与表面准备评估金属兼容性。
- `.agents/skills/xmake-plastichubs-website-ops/references/manufacturing-claims-control.md`：限定能力、认证、精确指标和受监管用途表述。
- Internal — Not for Publication：FESPA Screen Printing Fundamentals：支持网版、刮板、油墨、基材及表面润湿关系的基本工艺说明。
- Internal — Not for Publication：Marabu Mara Glass MGL Technical Data Sheet：支持金属、薄层阳极氧化铝、镀铬及涂装表面需要清洁、预试和油墨体系匹配。
- Internal — Not for Publication：Nazdar Ink Selection for Screen Printing on Metal：支持金属实际涂层、表面化学、油墨润湿、固化和附着测试共同决定结果。
- Internal — Not for Publication：ASTM D3359-23：支持金属基材上较有延展性涂层的胶带附着分级，并明确其不是绝对剥离强度测量。
- Internal — Not for Publication：ISO 2409:2020：支持通过划格评估涂层抗分离性，并要求按涂层和底材选择适用步骤。
- 本稿未承诺固定最小线宽、字符尺寸、套准、油墨厚度、颜色范围、耐磨循环、户外寿命、交期或保证性结果。
- `.agents/skills/xmake-plastichubs-website-ops/references/surface-finishing-knowledge-base.md`: Supports screen printing for graphics, logos, labels, and panels on flat or gently curved surfaces and requires metal compatibility review around ink and surface preparation.
- `.agents/skills/xmake-plastichubs-website-ops/references/manufacturing-claims-control.md`: Controls capability, certification, exact metrics, and regulated-use language.

- Internal — Not for Publication: FESPA Screen Printing Fundamentals: Supports the basic process description covering the screen, squeegee, ink, substrate, and surface wetting.
- Internal — Not for Publication: Marabu Mara Glass MGL Technical Data Sheet: Supports cleaning, preliminary trials, and ink-system matching for metals, thinly anodized aluminum, chrome-plated parts, and coated surfaces.
- Internal — Not for Publication: Nazdar Ink Selection for Screen Printing on Metal: Supports the combined effects of actual metal coating, surface chemistry, ink wetting, cure, and adhesion testing.
- Internal — Not for Publication: ASTM D3359-23: Supports tape-test rating of relatively ductile coatings on metallic substrates and states that it is not an absolute peel-strength measurement.
- Internal — Not for Publication: ISO 2409:2020: Supports cross-cut assessment of coating resistance to separation with procedures selected around coating and substrate.
- This draft does not promise a fixed minimum line width, character size, registration, ink thickness, color range, abrasion cycles, outdoor life, lead time, or guaranteed result.